

Predicting Costa Rican Household Poverty Levels

Introduction

The UN's Sustainable Development number 1 objective for 2030 is to eliminate poverty worldwide (United Nations Statistics Division, 2023)¹. This has been the group's primary target for the past 9 years (since 2015). Their complete objective is composed of 17 different smaller goals over a 15-year timeline (2015–2030). With the achievement of all smaller goals leading to the end of global poverty in all its forms by 2030. Some of these smaller goals include clean water and sanitation, reduced inequalities, and sustainable cities and communities.

The World Bank's Poverty data estimated that roughly 150 countries still had some level of extreme poverty in 2022 (to varying extents)². Unfortunately, Costa Rica has been on this list since it's conception in 2015. While Costa Rica has remained one of the wealthier countries in Latin America and the Caribbean. The poverty rate has continued to hover around 20% to 25% since 2010 (World Bank Group, 2020). The 2020 COVID-19 crisis has also pushed the economy further down and poverty and inequality up. In comparison, Costa Rica has double the US's poverty rate at 11.1% in 2023 (The White House, 2024)³.

For our analysis, we decided to focus on how to help the UN's Sustainable Development goal by understanding poverty in Costa Rican Households. With a stable government and good economic outlook, Costa Rica has the opportunity to greatly lower their rate of poverty. One way the Costa Rican government aids in this change is through programs such as the Puente al Desarrollo (Bridge to Development)⁴ (Tico Times, 2015). Launched in 2015, the Puente al Desarrollo aims to lift families out of extreme poverty by consolidating various welfare programs into a unified system tailored to individual household needs. It targets families earning below a certain income threshold, providing them with monthly stipends and access to services like healthcare and education to break the cycle of poverty. The goal was to assist over 54,000 families by integrating efforts from multiple public agencies. However, simply looking at low-income families can miss large families or others that are considered "higher income" but

still at significant risk. We can save time and money by aiding at risk families before they hit the extreme level of property many face.

How does our analysis aid with this? The **goal of our analysis is to build a machine learning model to identify households in poverty and at risk of poverty. By identifying those in poverty** (looking at more than income), **we can help programs such as the Puente al Desarrollo (Bridge to Development) better target families in need. We will achieve this by building a model that predicts poverty level with over 90% precision, based on income along with several other factors.** Such as home type, roof type, family size, number of dependents, etc. By using measures other than income, we can ensure that the program (and others like it) targets all of those who are in need of support in Costa Rica. In practice, the government employees at Puente al Desarrollo, can use our model to input household data, and assist those who are classified as poverty or at risk.

Understanding the nuances of household poverty can lead to more effective use of government resources, directly impacting the lives of families by reducing poverty, enhancing living standards, and potentially breaking the cycle of poverty. By predicting which households are most at risk, policymakers can prioritize interventions where they are most needed, ensuring that aid reaches those who will benefit most from it. For stakeholders like government agencies, NGOs, and community leaders, this project offers a strategic approach to poverty alleviation. It's not just about distributing aid but doing so with precision, thereby maximizing the impact of every dollar spent. This model could be pitched as a tool for sustainable development, offering a data-driven method to understand and address the root causes of poverty at the household level.

Our data was gathered from a comprehensive household survey of 9,557 households conducted by the Instituto Nacional de Estadística y Censos (INEC) of Costa Rica in July of 2023. Supplemented by demographic and socio-economic indicators from the World Bank. It consisted of 143 questions and measures on various items such as household size, number of dependents, education level, roof material, number of tablets in home, method of trash removal, and property level etc. Poverty level, being our predictor variable.

- 1 = Extreme Poverty
- 2 = Moderate Poverty
 - 3 = Vulnerable
- 4 = Non-Vulnerable

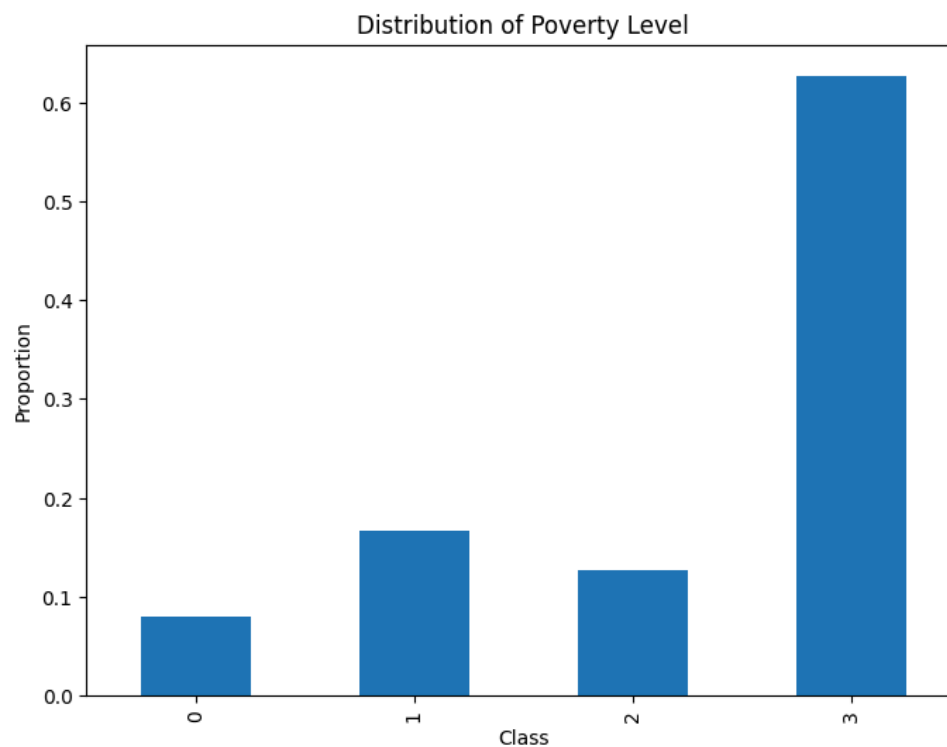
Data preparation & EDA

The first step we took to prepare our data was renaming columns for clarity and better usability. As many of our features contained names such as “v14a”, making EDA more difficult. Due to the vast number of measures, missing data was to be expected. We found two variables with over 90% missing values, which were removed before the training of our model.

'Monthly_rent' and 'Years behind in school' had over 40% missing values, so they were also dropped from our dataset. Any other missing values (10 total) were filled in with the medium. Duplicate variables were also removed. Variables causing multicollinearity were looked at a bit closer before determining which to remove (removed to prevent overfitting in our model). We reviewed the R values of any variables with an R over absolute value of 0.8. After review, we determined that 9 variables were highly correlated due to their similarity (e.g. tablet (yes/no), number of tablets) removed the one of less significance (e.g. removed “tablet”, keeping the exact number of tablets per household in the data set). The removal of such variables resulted in a total of 131 to feed our final model. There were also a small number of variables with text in place of a number value. Such as “number of dependents”. Where "no" meaning that there are no dependents and "yes" there is at least 1. To fix this we replaced all "no"s with 0 and "yes"s with 1. This was done for three measures.

Before we even began removing the variables listed earlier, we performed exploratory analysis (EDA) to further understand our data set and determine what type of data model would be most appropriate based on our data and goal. Through our EDA, valuable insights were found through a number of different visuals. We observed the variables causing multicollinearity through correlation matrices and heat maps. The correlation matrix on the number of children in a household vs poverty level showed us that it is NOT highly correlated to poverty level like one would expect ($R = -0.32$). Histograms were able to show us the

distribution in age and the number of households per poverty level. Since poverty level is our target/predictor variable, it was very important to have an equal distribution for our model to be accurate. This histogram showed a very skewed distribution, with most households labeled as “4, Non-Vulnerable”.

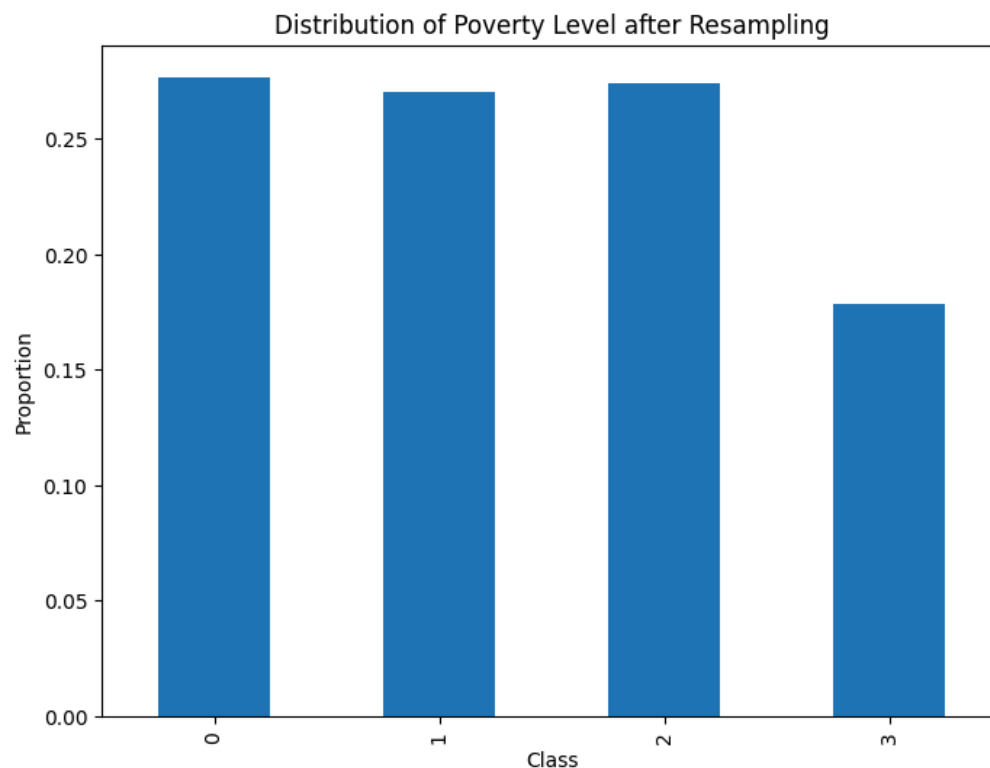


This right skew will significantly impact our model if not addressed. Luckily this was easily corrected by using a combination of oversampling and undersampling.

- **Oversampling** - Synthetic Minority Over-sampling (SMOTE) to generate synthetic samples for minority classes (Class 0, 1, 2).
- **Undersampling** - Reduce the number of samples from the majority class (Class 3) to balance the dataset.

We have chosen this combination method, since the observations that fall under the minority classes (0, 1, 2) are the ones we need to identify in our model. Having less of the non-risk class

(3) should not impact our project's goal. We also chose not to go the route of adjusting the model's class weights due to how it penalizes misclassifications of minority classes more heavily.



Model Building & Valuation

In order to choose a model, we first identify what type of data we are working with and if we are making a binary prediction (true/false), continuous, or ranked. Since we are looking to predict the poverty level between 1 and 4 (1 being extreme poverty and 4 being not vulnerable) we will need to choose an ordinal model (ordinal classification problem). As the order of our prediction variable is important.

Now this gives us a good selection of models to choose from. Some common models for ordinal data includes Ordinal Regression models (Logistic, Probit), Tree-Based Models (XGBoost, LightGBM, CatBoost) and Neural Networks. Each model has its own strengths and weaknesses. It is important to understand each, as one may be better at large data sets, one may be easier to interpret (if that is a necessity), or another could be powerful yet requires feature scaling and careful tuning.

After carefully reviewing our data and the strength and weakness of each model for an ordinal classification problem, we selected a Gradient Boosting Model (XGBoost). We chose this model due to its ability to handle mixed feature types well. As we have a combination of binary and continuous predictor variables. Gradient Boosting Models are also great at handling non-linear relationships and can be tuned for class imbalance if needed. The downside is that it is not as easy to interpret as an Ordinal Regression or Tree-Based model. But since we do not need to fully understand the model's inner workings to achieve our goal of predicting poverty, we will test out our data on the Gradient Boosting Model first.

Our first Gradient Boosting Model (XGBoost) model contained the following parameters:

- **objective = 'multi:softmax'** – parameter tells the model that this is a multi-class classification problem (predicting one of the 4 classes of property). multi:softmax means the model will output a single poverty level (1, 2, 3, or 4) by picking the class with the highest probability.
- **num_class = 4** – **Specifies** there are 4 classes to predict (poverty levels 1 through 4).
- **eval_metric = 'mlogloss'** – Sets the evaluation metric to multi-class log loss. This measures how well the model's predicted probabilities match the true labels. It helps the model focus on getting probabilities right, even though softmax outputs a single label.
- **random_state = 42** – Random seed to aid with reproducibility.

This model was trained on our training set (once) and tested against our test set. While cross validation should always be done, we wanted to see how the model performed with one training iteration. So how did it perform? Pretty well!

- **MAE: 0.1156**
- **QWK: 0.8960**

The Mean Absolute Error (MAE) measures the average absolute difference between predicted and true poverty levels. For our ordinal problem with classes 0 to 3 (after shifting from 1-4 to 0-3), MAE ranges from 0 (perfect predictions) to 3 (worst-case predictions), meaning an MAE of 0.1156 is near perfection! The exact meaning is that the model's predictions are off by about 0.12 levels. Indicating that most predictions are either correct or only off by only 1 class/poverty level. This is great news and means the model is performing very well in terms of minimizing prediction errors.

The Quadratic Weighted Kappa (QWK) measures agreement between predicted and true classes, with quadratic weighting to penalize larger errors more heavily. It is particularly suitable for ordinal classification because it accounts for the ordered nature of the classes. QWK ranges from -1 (complete disagreement) to 1 (perfect agreement), with 0 indicating random agreement. Our high result of QWK = 0.8960, indicates a strong agreement between predicted and true poverty levels. According to common guidelines a QWK between 0.81–1.00 is almost perfect.

Overall, the low MAE and high QWK shows us that the model is performing very well. When we take a look at Class-Specific Performance, we see a similar trend as well as some small problems. There was high recall and precision for all Classes. However, Class 2 (poverty level 3) was slightly lower on recall and Class 1 (poverty level 2) for Precision. Meaning that the model may misclassify poverty level 3 and 2 some of the time. observation correctly with some minor issues in Class 2. Misclassifying some 2s and 1s as 3s in some cases.

Even though we had good results on our first model, we want to use 5-fold cross validation along with Gridsearch to see if we can improve this metric for poverty level 2 and 3. GridSearch will try several combinations of the parameters passed by the dictionary and evaluate the best model for each combination. 5-fold cross validation will then be applied to each fit, splitting and resampling the training data for each of the models (5 different portions of the training set). GridSearchCV works with 5-fold cross validation by testing several combinations and evaluating the best model for each combination with Cross-Validation.

We set our dictionary for the GridSearchCV with the following parameters,

- **max_depth: [3, 5, 7]** – Maximum depth of each decision tree in the XGBoost model. Depth being how many levels of splits a tree can have. 3, 5, 7 were chosen to give a variety of depth without the risk of overfitting (more depth).
- **learning_rate: [0.01, 0.1, 0.3]** – How much each tree contributes to the final prediction. Smaller values are less likely to overfit while larger values fit faster but overfitting. Our options lean on the smaller side to avoid overfitting.
- **n_estimators: [100, 200, 300]** – The number of decision trees XGBoost builds.
- Fewer trees may underfit complex data (missing nuanced poverty patterns). More trees can be slower to train, and risk overfitting.
- **subsample: [0.8, 1.0]** – Fraction of training data randomly sampled for each tree. E.g. 0.8 = 80%. The default was chosen.
- **colsample_bytree: [0.8, 1.0]** – Fraction of features randomly sampled for each tree.

How did our GridsearchCV perform compared to our standard/first fit? Very well! The MAE was even lower than before. Going from 0.1156 to 0.0842. The QWK also indicates an improved model with a higher value (0.9241) than before (0.8960). This improvement shows just how much Gridsearch and Cross Validation can improve a model.

- **MAE: 0.0842**
- **QWK: 0.9241**

We also saw an improvement in our Class-Specific Performance when comparing the new model to our first. There was a slight improvement in precision for poverty level 3, meaning that when the model predicts level 3, it's more likely to be correct. However, there is a noticeable decline in recall for poverty level 2, which could be an issue, as identifying instances at this poverty level (at risk) is critical. Poverty level 1 and 4 show no change to their previous performance levels.

To summarize our model's performance, the combination of oversampling and undersampling to smooth the distribution of poverty levels was a necessary step prior to

building our model. This enabled even the basic XGBoost model (before GridSearchCV) to perform very well. However, the XGBoost model chosen by GridSearchCV performed the best and will be model chosen for implantation. The chosen model contained the below key parameters,

- objective = 'multi:softmax'
- 'learning_rate': 0.3
- 'max_depth' = 7
- num_class = 4
- eval_metric = 'mlogloss'
- 'n_estimators' = 200
- 'subsample': 1.0
- 'colsample_bytree': 0.8

Conclusion

While best model chosen by GridSearchCV performed well, we are still concerned about the decline in recall for Poverty Level 2, as it is one of the most important classes for our prediction of poverty in Costa Rica. We will need to continue to review ways to improve this aspect of the model prior to implementation. Possibly by adding weights or trying other models for ordinal classification problems.

Even if our model is not quite ready for deployment, our initial analysis has revealed that it is very possible to predict household poverty levels in Costa Rica based on measurable factors such as family size, roof type, type of trash removal, education level, etc. This shows that a solid model can aid poverty reduction programs in Costa Rica. By enabling teams to enter survey data to obtain a predicted level of poverty. Helping target those who are predicted to be most at risk for poverty and helping reduce any bias from using only income as an indicator (as it is done today). Once the final model is ready, we recommend integrating it into the social welfare system of Costa Rica to enhance the targeting of support programs.

Before we come to a final conclusion, it is important to highlight some of the challenges that could arise after model implantation. Some of these challenges include data privacy, adapting

the model to new demographic trends, and integrating feedback from real-world use into model updates. We can preemptively address these problems by removing any personal data from new entries before adding them to the model, bi-yearly/as needed model re-training, and add new observations/household feedback to the model before re-training.

Some of the additional opportunities include expanding on the data feeding our model. Adding measures for specific areas, such as local policies, price of goods, and community health. There is also potential to link this model with employment databases for a more holistic approach to poverty alleviation.

Overall we are excited about our initial models performance (92% correct predictions (QWK)). And look forward to making the final adjustments to address the accuracy of poverty level 2 predictions before putting our model to use. Helping Costa Rica address poverty more efficiently and ultimately aid the UN's Sustainable Development goal of to eliminating poverty worldwide by 2030.

References on next page

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